

FreshRoute Agent

Product Requirements Document (PRD)

Document field	Details
Product name	FreshRoute Agent
Product category	AI-powered agricultural market intelligence, produce-quality assessment, logistics coordination, and transaction-execution platform
Primary market	Pakistan, beginning with high-volume horticulture and perishable-produce corridors
Primary users	Farmers, produce traders, commission agents, collection-center operators, buyers, transporters, cold-storage providers, operations teams
Platforms	Mobile web, Android application, WhatsApp, admin web portal, partner portal, APIs
Initial language support	Urdu, Roman Urdu, English
Product stage	MVP → Pilot → Marketplace → Intelligence platform
Core value proposition	Help produce sellers decide the best market, timing, buyer, transport, and storage option—then execute and monitor the sale workflow
Product principle	“Do not just recommend. Analyze → compare → contact → book → monitor → alert.”

1. Executive Summary

FreshRoute Agent is an AI-powered operating system for fresh-produce selling and post-harvest coordination. A farmer, trader, or collection-center worker can upload a produce photo, send a WhatsApp message, share a voice note, or use the application to describe an available lot.

For example:

“I have 800 kg tomatoes in Multan. They will be ready tomorrow.”

FreshRoute Agent converts that unstructured message into a structured produce lot, analyzes visible quality, checks market price data, estimates spoilage risk,

finds matching buyers and logistics providers, compares sell-now versus store-and-sell-later outcomes, and recommends an execution plan.

After user approval, the platform sends buyer inquiries, requests transporter quotations, reserves storage, tracks delivery, monitors changing prices and conditions, and alerts the user if a better option or an operational issue appears.

The product addresses a meaningful economic problem in Pakistan: post-harvest losses associated with weak storage, transport, cold-chain, and market-linkage systems are estimated in the billions of dollars annually. Some studies also estimate that 30–40% of fruit and vegetable output can be lost across the value chain. breccorder

2. Problem Statement

2.1 Current User Problems

Farmers and produce traders frequently make selling decisions using incomplete, delayed, or informal information.

They often do not know:

- Which market currently offers the best realizable price.
- Whether the reported market price is actually achievable after commission, transport, loading, and wastage.
- Whether to sell immediately, wait, store, or move produce to another market.
- Whether their produce quality qualifies for a premium buyer.
- How much produce is likely to spoil during waiting, transport, or storage.
- Which transporter has a suitable available vehicle.
- Whether refrigerated transport or cold storage is required and affordable.
- What buyer documentation, grade standards, packaging, or delivery windows are required.
- How to coordinate farmers, traders, buyers, drivers, storage operators, and laborers over fragmented WhatsApp messages and phone calls.
- Whether a booked buyer or transporter has changed terms, delayed, cancelled, or failed to respond.

2.2 Current Workflow Problems

The current workflow is fragmented:

Farmer harvests produce

↓

Calls local trader or mandi commission agent

↓

Gets informal verbal prices
↓
Searches manually for truck/loader/storage
↓
Negotiates repeatedly on phone or WhatsApp
↓
Moves produce with uncertain demand and pricing
↓
Faces spoilage, rejection, delays, or forced low-price sale

The result is usually one or more of the following:

- Lower realized price.
- Product spoilage.
- Buyer rejection because of grade mismatch.
- Higher transport cost.
- Delayed payment.
- Unused storage capacity.
- Wasted time in coordination.
- No reliable transaction record.
- No data trail to improve future decisions.

2.3 Opportunity

FreshRoute Agent creates value by turning disconnected agricultural decisions into a data-supported and operationally executable workflow.

The platform is not merely a crop-advisory chatbot. It acts as a controlled transaction agent that can:

1. Understand the produce lot.
2. Estimate quality and risk.
3. Find the economically best route.
4. Contact suitable counterparties.
5. Secure approvals.
6. Book logistics and storage.
7. Track the execution.
8. Alert users to changes or exceptions.

3. Product Vision

3.1 Vision Statement

Enable every farmer and produce trader to sell perishable produce through an intelligent, trusted, and execution-capable digital agent.

3.2 Mission

Reduce post-harvest losses, improve realized selling prices, and make agricultural logistics and buyer access more transparent for Pakistan’s produce ecosystem.

3.3 Product Principles

- **Action over information:** Recommendations must lead directly to executable actions.
 - **Human approval for commitments:** The system must never sell, book, or commit funds without explicit user authorization.
 - **Explainable recommendations:** Users must see why one option is recommended over another.
 - **Low-friction interaction:** The product must work for users who are comfortable with WhatsApp, voice notes, and photos rather than complex forms.
 - **Local context first:** Support Urdu, Roman Urdu, mandi-specific practices, local units, crop grades, and regional logistics realities.
 - **Trust and auditability:** Every offer, approval, booking, and status change must be recorded.
 - **Graceful degradation:** When live data is unavailable, the product should clearly label estimates, cached data, or manually sourced information.
-

4. Product Goals

4.1 Primary Goals

- Improve the net realized revenue per produce lot.
- Reduce avoidable produce spoilage.
- Reduce time spent coordinating buyers, transporters, and storage providers.
- Increase access to reliable market and logistics information.
- Improve the fill rate of buyer demand and logistics capacity.
- Create a verified data network of produce lots, buyers, transporters, cold storages, and completed transactions.

4.2 Secondary Goals

- Help farmers meet buyer quality requirements.
- Improve utilization of cold storage and transport capacity.
- Generate regional price and supply intelligence.
- Build a trusted marketplace for agricultural trade and logistics.
- Create data infrastructure for future offerings such as financing, insurance, certification, and export facilitation.

4.3 Non-Goals for MVP

The initial MVP will not:

- Guarantee market prices.
 - Guarantee produce quality, yield, or buyer acceptance.
 - Replace physical inspection for premium/export-grade purchases.
 - Provide lending or insurance underwriting.
 - Process direct buyer-to-farmer payments as the primary transaction method.
 - Conduct autonomous price negotiation without explicit user-defined rules.
 - Provide legal certification itself.
 - Serve every crop from day one.
-

5. Target Users and Personas

5.1 Farmer

Profile

A small or medium farmer with 200 kg to 20,000 kg of produce, often selling through local traders, mandis, aggregators, or collection centers.

Needs

- A simple way to learn where and when to sell.
- Better price visibility.
- Confidence that transport is available.
- Help understanding quality and buyer requirements.
- A workflow that works through WhatsApp and voice messages.

Pain Points

- Limited access to multiple markets.
 - Information asymmetry.
 - Urgent need to sell perishable produce.
 - Lack of cold storage.
 - Dependence on local intermediaries.
 - Limited comfort with complex digital applications.
-

5.2 Produce Trader / Wholesaler

Profile

A trader handling frequent lots from farms and selling to mandis, retailers, restaurants, processors, and distributors.

Needs

- Fast buyer matching.
- Lot-level profitability comparison.
- Centralized deal and logistics tracking.
- Better forecasting of supply, demand, and price movement.
- Ability to coordinate multiple workers.

Pain Points

- High volume of calls and WhatsApp messages.
 - Missed buyer inquiries.
 - Weak traceability.
 - Delayed transport confirmation.
 - Difficulty tracking partial loads and delivery status.
-

5.3 Collection-Center Operator

Profile

An organization collecting produce from multiple farms, grading, consolidating, and selling in bulk.

Needs

- Lot intake and quality classification.
- Aggregation planning.
- Storage and dispatch coordination.
- Buyer demand matching.
- Staff workflow management.

Pain Points

- Inconsistent quality records.
 - Manual inventory tracking.
 - Lack of buyer-side demand visibility.
 - Difficulty forecasting packing and dispatch needs.
-

5.4 Buyer / Retailer / Processor

Profile

A supermarket, restaurant group, wholesaler, food processor, exporter, or distributor buying defined quantities with quality and delivery requirements.

Needs

- Reliable supply.
- Standardized quality information.
- Timely delivery.
- Transparent pricing.
- Supplier reliability.

Pain Points

- Inconsistent quality.
 - Last-minute shortages.
 - Lack of traceability.
 - Long supplier coordination cycles.
 - Limited visibility into upcoming supply.
-

5.5 Transporter

Profile

A truck owner, driver, logistics broker, refrigerated-vehicle operator, or small fleet manager.

Needs

- More predictable load demand.
 - Better route utilization.
 - Fewer empty return trips.
 - Digital proof of booking and completion.
 - Clear pickup and delivery instructions.
-

5.6 Cold-Storage Provider

Profile

A cold-room owner or warehouse operator with perishable storage capacity.

Needs

- Increase capacity utilization.
 - Receive structured booking requests.
 - Know crop, quantity, temperature range, duration, and arrival time.
 - Reduce phone-based booking confusion.
-

6. User Stories

6.1 Farmer User Stories

- As a farmer, I want to upload a photo of my tomatoes so the system can identify the crop and visible quality.
- As a farmer, I want to send a voice note in Urdu instead of completing a long form.
- As a farmer, I want to know whether I should sell today or store until tomorrow.
- As a farmer, I want to compare net earnings across different markets after transport and spoilage costs.
- As a farmer, I want the system to find an available transporter.
- As a farmer, I want to approve a buyer message before it is sent.
- As a farmer, I want immediate alerts if my buyer cancels or my vehicle is delayed.

6.2 Trader User Stories

- As a trader, I want to create multiple produce lots and see their recommended markets.
- As a trader, I want to compare transporters by price, availability, vehicle type, and reliability.
- As a trader, I want to message multiple buyers from one dashboard.
- As a trader, I want to see my expected margin before confirming a transaction.
- As a trader, I want to track all active lots, bookings, payments, and deliveries.

6.3 Buyer User Stories

- As a buyer, I want to define procurement requirements for crop, grade, quantity, location, and delivery window.
- As a buyer, I want to receive only relevant supply offers.
- As a buyer, I want photos, estimated grade, quantity, and expected arrival time before accepting an offer.
- As a buyer, I want to confirm, reject, or counteroffer through WhatsApp or the portal.

6.4 Transporter User Stories

- As a transporter, I want to receive load opportunities matching my vehicle and route.
 - As a transporter, I want pickup and destination details before accepting a booking.
 - As a transporter, I want to update arrival, pickup, in-transit, delay, and delivered status.
 - As a transporter, I want digital proof of delivery.
-

7. Product Scope

7.1 MVP Crop Scope

The MVP should launch with a limited set of high-volume and highly perishable crops:

- Tomatoes.
- Potatoes.
- Onions.
- Mangoes.
- Kinnow.
- Bananas.
- Green chilies.
- Okra.
- Leafy vegetables.

Tomatoes should be the flagship crop because quality variability, short shelf life, price volatility, and frequent transport needs make the value proposition obvious.

7.2 MVP Geography

Recommended launch geography:

- Multan.
- Lahore.
- Faisalabad.
- Islamabad/Rawalpindi.
- Karachi.
- Surrounding farm-to-mandi corridors.

The first pilot should focus on one crop corridor, such as:

Multan farms

↓

Multan mandi / collection center

↓
Lahore / Islamabad / Karachi buyers

7.3 Channel Scope

- Android application.
 - Mobile web application.
 - WhatsApp Business integration.
 - Admin portal.
 - Buyer portal.
 - Transporter portal.
 - Cold-storage partner portal.
-

8. Core Product Modules

Module	Purpose
Produce Lot Intake	Capture crop, quantity, location, timing, images, quality, and seller intent
AI Vision Analysis	Identify crop and estimate visible quality indicators
Conversational Agent	Ask clarifying questions and guide user decisions
Market Intelligence	Retrieve market price, demand, historical trends, and buyer offers
Spoilage Risk Engine	Estimate likely loss from waiting, heat, handling, transport, and storage
Recommendation Engine	Compare sale, storage, transport, and buyer options
Buyer Marketplace	Match lots with qualified buyers
Logistics Marketplace	Match lots with transporters and storage providers
Workflow Orchestrator	Execute approved messages, bookings, reminders, and monitoring jobs
Notifications Engine	Send WhatsApp, SMS, push, and call-escalation alerts
Order Tracking	Track lot status from availability through payment and delivery
Admin and Operations Portal	Monitor users, quality, disputes, exceptions, and marketplace operations

Module	Purpose
Analytics and Reporting	Report prices, performance, spoilage, revenue, fill rates, and operational metrics

9. End-to-End Application Workflow

9.1 Primary Workflow

```

User submits photo / voice note / text
↓
FreshRoute ingests and validates the request
↓
AI extracts crop, quantity, location, readiness date, and visible quality
↓
Agent asks only required clarifying questions
↓
System creates a structured produce lot
↓
Market, buyer, transport, weather, and storage data are retrieved
↓
Spoilage and revenue scenarios are calculated
↓
Agent creates ranked action recommendations
↓
User selects an action or asks the agent to execute
↓
Agent drafts messages / booking requests
↓
User explicitly approves the outbound action
↓
System contacts buyers, transporters, and storage providers
↓
Responses are captured and ranked
↓
User approves final counterparty and commercial terms
↓
Booking and order records are created
↓
Agent monitors delivery, delays, buyer changes, and pricing changes
↓
User receives alerts and exception recommendations

```

↓
Delivery is confirmed and transaction is closed
↓
User is prompted for outcome feedback and quality verification

9.2 Example: Tomato Selling Flow

Input

User message:

"I have 800 kg tomatoes in Multan. They will be ready tomorrow."

Attachments:

3 tomato photos

Optional voice note

Step 1: Agent Extraction

The system extracts:

```
{  
  "crop": "tomato",  
  "quantity_kg": 800,  
  "location_city": "Multan",  
  "harvest_ready_date": "2026-08-30",  
  "seller_type": "farmer",  
  "confidence": {  
    "crop": 0.96,  
    "quantity": 0.87,  
    "location": 0.92,  
    "harvest_date": 0.91  
  }  
}
```

Step 2: Vision Analysis

The vision model analyzes uploaded photos for:

- Crop type.
- Ripeness stage.
- Visible bruising.
- Surface defects.
- Color consistency.
- Size variation.
- Packaging condition.
- Presence of mold or visible disease symptoms.

- Approximate grade confidence.

Gemini's multimodal capabilities can perform image understanding, image classification, visual question answering, and object detection workflows, making it appropriate for initial produce-image interpretation. business.whatsapp

Example result:

```
{
  "crop_type": "tomato",
  "quality_grade_estimate": "B",
  "ripeness": "medium-high",
  "visible_defect_rate_estimate": 0.08,
  "packaging_condition": "unconfirmed",
  "confidence_score": 0.74,
  "human_review_required": false
}
```

Step 3: Clarifying Questions

The agent should ask only questions that materially affect the recommendation.

Example WhatsApp exchange:

FreshRoute:

I found tomatoes with medium-to-high ripeness. To recommend the best option, please confirm

1. Are they packed in crates, sacks, or loose?
2. Do you have shade or cold storage available overnight?
3. Is 800 kg the final expected quantity?
4. Can the load leave Multan tomorrow morning?

Step 4: Data Enrichment

The system retrieves:

- Mandi market prices.
- Buyer demand and open requests.
- Buyer grade requirements.
- Transporter capacity and quotes.
- Cold-storage availability and cost.
- Distance and travel time.
- Current and forecast weather.
- Historical price movement.
- Crop-specific spoilage coefficients.
- Seller reliability and previous transaction history.

Step 5: Financial Scenario Comparison

The system produces scenarios.

Scenario	Market / Buyer	Gross Revenue	Transport	Storage	Estimated Spoilage	Commission	Expected Net Revenue	Risk
Sell today	Local Multan mandi	PKR X	PKR X	PKR 0	3%	PKR X	PKR X	Low
Sell tomorrow	Lahore wholesale buyer	PKR X	PKR X	PKR 0	8%	PKR X	PKR X	Medium
Store one day	Multan cold storage + buyer	PKR X	PKR X	PKR X	4%	PKR X	PKR X	Medium
Premium buyer	Retail buyer in Lahore	PKR X	PKR X	PKR X	6%	PKR X	PKR X	Medium-High

Step 6: Recommendation

Example response:

Recommended action: Sell tomorrow morning to Buyer A in Lahore.

Why:

- Expected net revenue is estimated to be 13% higher than selling locally.
- The buyer accepts Grade B tomatoes.
- Refrigerated transport is not essential for the estimated 7-hour route, but crate packaging is.
- Storing for two days is not recommended because spoilage risk increases materially.
- Buyer A has an 82% historical acceptance rate and can take the full 800 kg lot.

Step 7: Execution Approval

FreshRoute:

Would you like me to:

1. Send an offer to Buyer A for 800 kg?
2. Request a quote from 3 transporters?
3. Reserve 1 day of cold storage as backup?

No message or booking will be sent without your approval.

Step 8: User Approval

User:

Yes, contact Buyer A and ask transporters.

Step 9: Agent Action

The orchestration engine:

- Creates buyer inquiry tasks.
- Sends WhatsApp message templates.
- Requests transporter availability.
- Starts response timeout timers.
- Records all outbound communication.
- Updates the lot timeline.

Step 10: Confirmation and Booking

Once buyer and transporter respond:

FreshRoute:

Buyer A accepted 800 kg at PKR X/kg.

Transporter B can collect at 7:00 AM for PKR X.

Estimated arrival: 2:30 PM.

Expected net revenue: PKR X.

Approve this sale and transport booking?

The user approves. Only then does the platform create the confirmed order and booking.

Step 11: Monitoring

The system monitors:

- Driver pickup status.
- Vehicle location if GPS is available.
- Temperature if cold-chain sensors are available.
- Buyer delivery window.
- Traffic and route disruption.
- Buyer cancellation signals.
- Market-price movement.
- Delay risk.
- Delivery confirmation.

Step 12: Completion

The delivery is closed once:

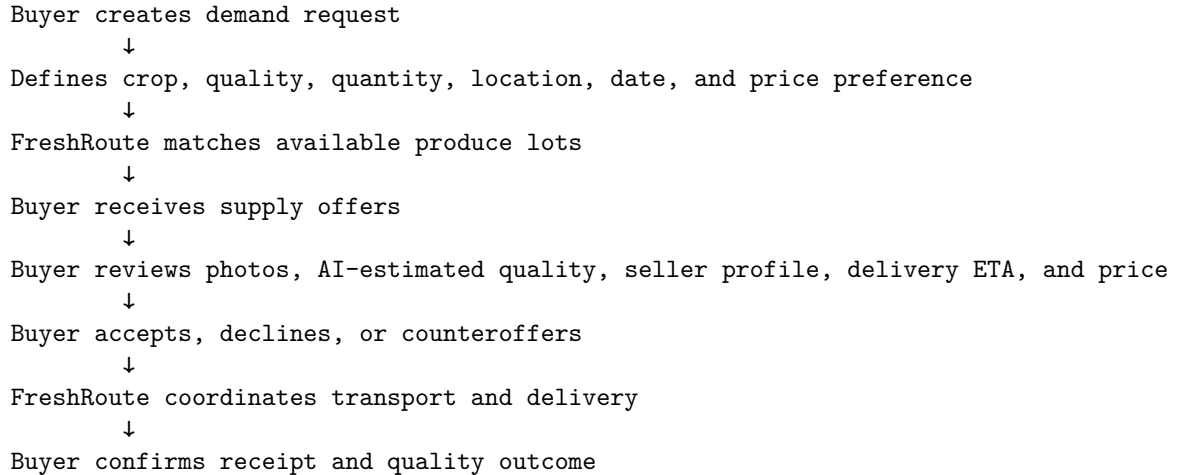
- Buyer confirms receipt.
 - Quantity is verified.
 - Quality acceptance or rejection is recorded.
 - Final sale price is recorded.
 - Payment status is updated.
 - User feedback is captured.
-

10. User Flow

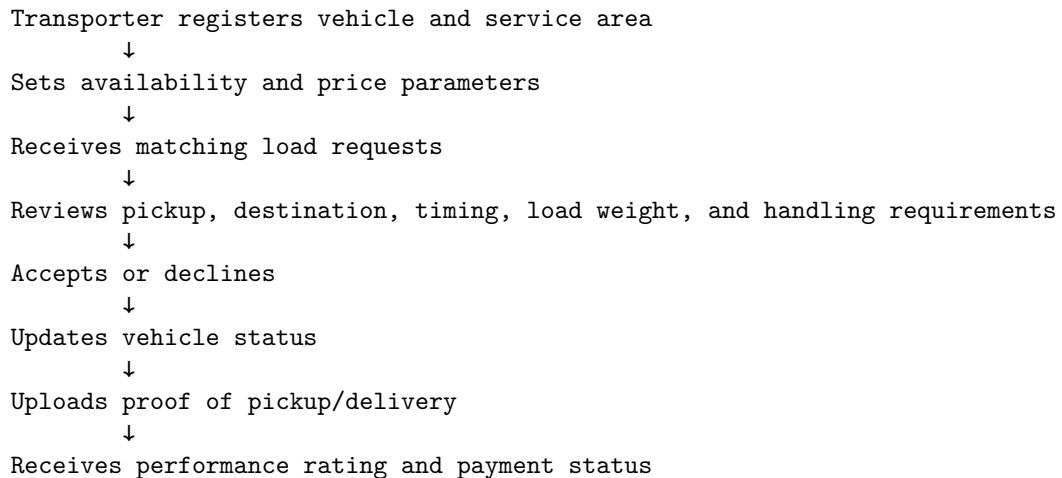
10.1 Farmer App User Flow

Open FreshRoute App / WhatsApp
↓
Tap "Sell Produce"
↓
Upload photo or record voice note
↓
Enter or dictate produce information
↓
AI extracts lot details
↓
Confirm details / answer clarifying questions
↓
View recommendation options
↓
Select "Sell Now," "Store," "Find Buyer," or "Compare Markets"
↓
Review expected earnings and risk
↓
Approve buyer outreach / transport quote / storage booking
↓
Review responses
↓
Approve final transaction
↓
Track pickup and delivery
↓
Confirm outcome and payment

10.2 Buyer User Flow



10.3 Transporter User Flow



11. Agentic Workflow Design

11.1 Agent Responsibilities

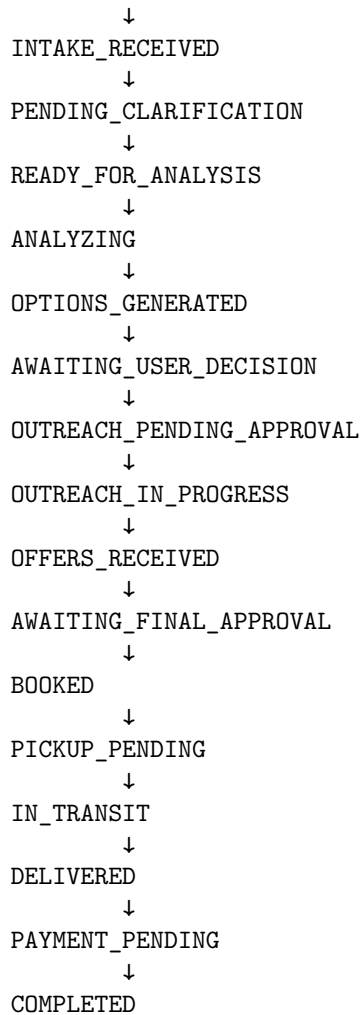
The FreshRoute Agent must be able to:

- Interpret natural-language requests.
- Process images and optional video.
- Extract structured lot data.

- Identify missing critical data.
- Ask concise clarification questions.
- Call approved internal and external tools.
- Generate financially grounded recommendations.
- Explain tradeoffs.
- Request user approval before irreversible actions.
- Track long-running workflows.
- Detect exceptions.
- Re-plan when conditions change.
- Escalate to human operations when confidence or risk is low.

11.2 Agent State Machine

DRAFT



Exception states:

LOW_CONFIDENCE
BUYER_CANCELLED
TRANSPORT_DELAYED
STORAGE_UNAVAILABLE
QUALITY_DISPUTE
PAYMENT_DISPUTE
HUMAN_REVIEW_REQUIRED
CANCELLED

11.3 Agent Decision Framework

The agent should evaluate the following variables:

Expected Net Revenue

=

(Expected Selling Price × Accepted Quantity)

- Transport Cost
- Storage Cost
- Commission
- Packaging Cost
- Estimated Spoilage Cost
- Delay Penalty
- Other Handling Costs

The agent should rank options using a weighted decision score:

Recommendation Score

=

Revenue Weight

- + Buyer Reliability Weight
- + Logistics Reliability Weight
- + Quality Match Weight
- Spoilage Risk Weight
- Delay Risk Weight
- Payment Risk Weight

An example conceptual scoring function:

Final Score =

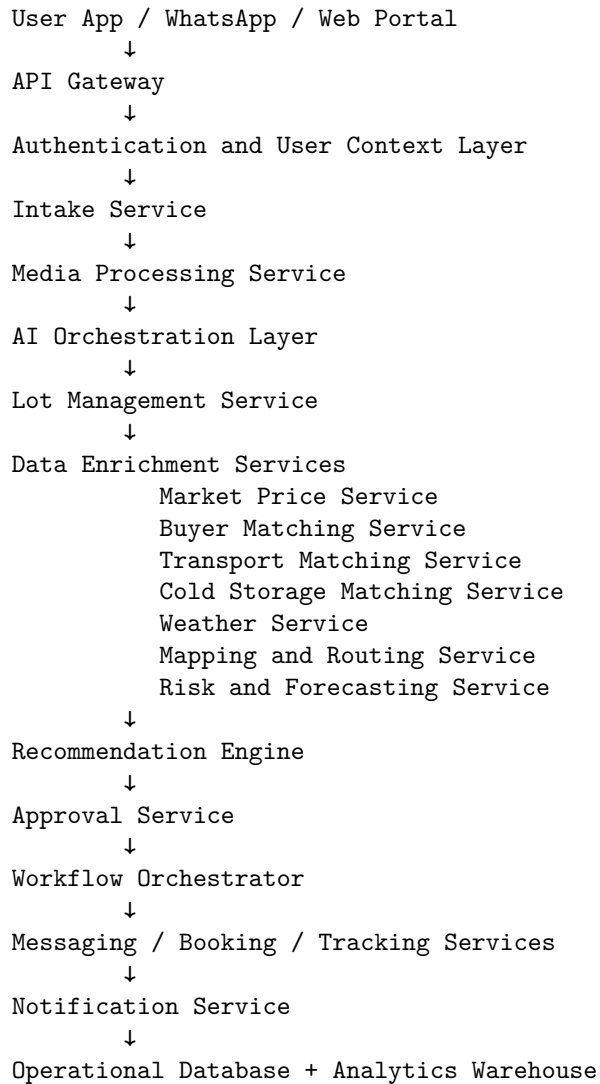
$0.40 \times \text{normalized_expected_net_revenue}$
 $+ 0.15 \times \text{buyer_reliability}$
 $+ 0.15 \times \text{logistics_reliability}$
 $+ 0.10 \times \text{quality_requirement_match}$
 $+ 0.10 \times \text{payment_terms_score}$
 $- 0.10 \times \text{spoilage_risk}$

Weights must be configurable by crop, user type, and user preferences.

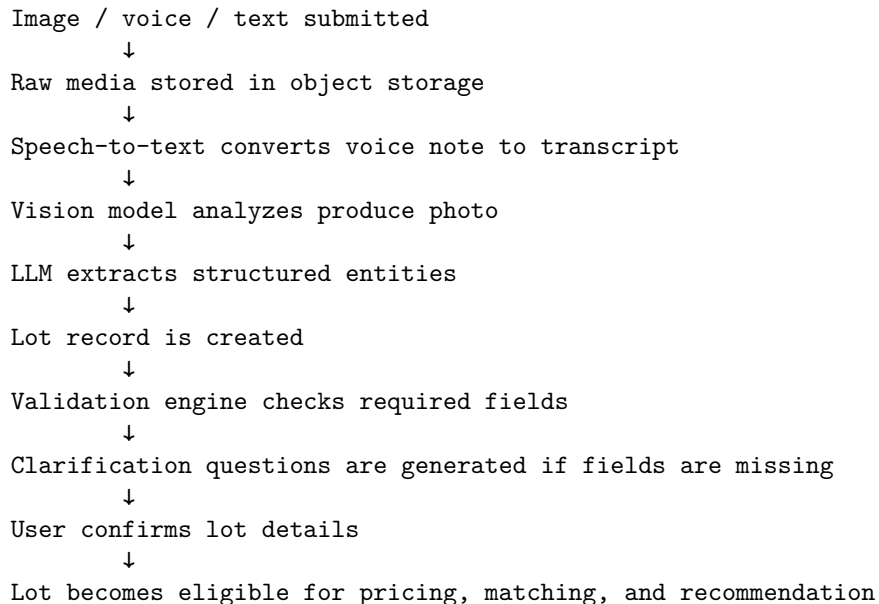
For example, a small farmer may prioritize cash payment and low risk, while a large collection center may prioritize margin and delivery capacity.

12. Data Flow Architecture

12.1 High-Level Data Flow



12.2 Produce Lot Data Flow



12.3 Market Data Flow



Market inputs may include:

- Official mandi price data where available.
- Partner uploads.
- Buyer offers.
- Trader-submitted transactions.
- Manual field-agent data.

- Historical FreshRoute completed sales.
- Public and private market data feeds.

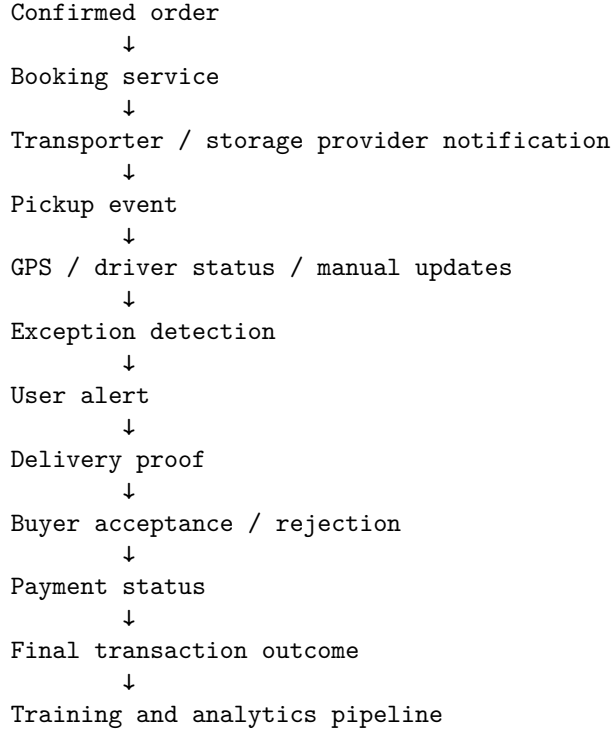
Every price should carry:

```
{
  "crop": "tomato",
  "grade": "B",
  "market": "Multan",
  "price_per_kg": 0,
  "currency": "PKR",
  "source": "market_feed",
  "observed_at": "timestamp",
  "confidence": 0.82,
  "data_freshness_minutes": 45
}
```

12.4 Recommendation Data Flow

```
Structured lot
+
Market price data
+
Buyer demand
+
Transport availability
+
Storage availability
+
Weather and temperature data
+
Historical spoilage data
+
Route time and cost
↓
Scenario engine
↓
Expected net revenue calculation
↓
Spoilage-risk prediction
↓
Counterparty ranking
↓
Recommendation generator
↓
User-facing action plan
```

12.5 Post-Booking Data Flow



13. Technical Architecture

13.1 Architecture Principles

- API-first.
- Event-driven for long-running workflows.
- Mobile-first and WhatsApp-native.
- Modular services with clear ownership.
- Secure-by-default handling of user and transaction data.
- Human-in-the-loop for high-impact actions.
- Observability across all AI and transactional workflows.
- Support for offline and low-bandwidth conditions.

13.2 Recommended Technology Stack

Layer	Recommended Technology
Mobile application	React Native or Flutter

Layer	Recommended Technology
Web application	Next.js / React
Backend APIs	Python FastAPI and/or Node.js with NestJS
AI orchestration	Python, LangGraph, Temporal workflows, custom tool orchestration
Vision and LLM	Google Gemini API for multimodal reasoning and image understanding
Speech-to-text	Google Speech-to-Text, Whisper, or Gemini audio capabilities where appropriate
Database	PostgreSQL
Geospatial queries	PostGIS
Cache and session store	Redis
Object storage	Google Cloud Storage, AWS S3, or Cloudflare R2
Vector search	pgvector, Pinecone, or Weaviate
Event streaming	Kafka, Google Pub/Sub, AWS SNS/SQS, or RabbitMQ
Workflow engine	Temporal
Search	Elasticsearch / OpenSearch
Analytics warehouse	BigQuery, ClickHouse, Snowflake, or PostgreSQL initially
Data transformation	dbt
Model training	Python, scikit-learn, XGBoost, PyTorch
MLOps	MLflow, Vertex AI, or SageMaker
Maps and routing	Google Maps Platform, Mapbox, or local logistics routing provider
Messaging	WhatsApp Business Platform, SMS gateway, push notifications
Authentication	Auth0, Firebase Auth, Clerk, or custom JWT/OAuth solution
Cloud infrastructure	Google Cloud Platform, AWS, or Azure
Containerization	Docker
Orchestration	Kubernetes, Cloud Run, ECS, or managed container service
Monitoring	Datadog, Grafana, Prometheus, Sentry, OpenTelemetry
CI/CD	GitHub Actions, GitLab CI, or CircleCI

WhatsApp Business Platform is suitable for sending and receiving user messages

programmatically and for operating business messaging at scale. breccorder

14. Backend Services

14.1 API Gateway

Responsibilities:

- Request routing.
- Authentication.
- API rate limits.
- Request logging.
- API version management.
- Security headers.
- Webhook validation.

14.2 Authentication Service

Responsibilities:

- Phone-number login.
- WhatsApp identity linking.
- OTP verification.
- Role-based access control.
- Session management.
- Consent management.
- Device registration.

Roles:

- Farmer.
- Trader.
- Buyer.
- Collection-center operator.
- Transporter.
- Storage provider.
- Field agent.
- Operations admin.
- Super admin.

14.3 Lot Management Service

Responsibilities:

- Create, update, and archive produce lots.
- Maintain lot state.
- Store quality assessment.

- Track ownership.
- Manage documents and images.
- Record inventory quantity changes.
- Link lots to buyer offers and orders.

14.4 AI Orchestration Service

Responsibilities:

- Receive user intent.
- Invoke speech, vision, LLM, forecast, and ranking models.
- Build structured prompts.
- Call internal tools.
- Track tool outputs.
- Generate explanations.
- Enforce user approval gates.
- Route low-confidence cases to human review.

14.5 Market Intelligence Service

Responsibilities:

- Ingest price feeds.
- Normalize crop names and grades.
- Maintain historical price time series.
- Calculate market trends.
- Produce price confidence scores.
- Serve market comparisons.

14.6 Spoilage Risk Service

Responsibilities:

- Estimate probability of spoilage.
- Calculate expected quantity loss.
- Explain top risk factors.
- Update risk as weather and timing change.
- Apply crop-specific deterioration models.

14.7 Matching Service

Responsibilities:

- Find buyers matching crop, grade, quantity, delivery window, and destination.
- Find transporters matching capacity, route, availability, vehicle type, and temperature needs.

- Find storage providers matching capacity, crop, temperature, and time duration.
- Rank counterparties by suitability and reliability.

14.8 Booking Service

Responsibilities:

- Create booking requests.
- Manage booking statuses.
- Prevent double booking.
- Store terms and conditions.
- Handle cancellations.
- Generate booking references.

14.9 Notification Service

Responsibilities:

- WhatsApp notifications.
- SMS fallback.
- Push notifications.
- Email for enterprise users.
- Voice-call escalation for severe exceptions.
- Template management.
- Delivery status tracking.

14.10 Order Tracking Service

Responsibilities:

- Pickup status.
- Loading status.
- In-transit status.
- Delay status.
- Delivery status.
- Quantity received.
- Quality acceptance.
- Payment status.
- Dispute status.

15. AI System Design

15.1 AI Components

AI Component	Purpose
Multimodal Vision Model	Crop recognition, visible quality analysis, defect detection, packaging assessment
Conversational LLM	Natural-language understanding, question generation, explanation, multilingual interaction
Entity Extraction Model	Convert unstructured text and voice into structured lot data
Spoilage Prediction Model	Predict expected quality loss and quantity loss
Price Forecasting Model	Estimate near-term price movement by market and crop
Buyer Ranking Model	Rank buyer opportunities by net revenue and likelihood of completion
Logistics Ranking Model	Rank transport and storage options by cost, feasibility, and reliability
Anomaly Detection Model	Detect suspicious prices, abnormal route delays, and unusual quality claims
Retrieval System	Retrieve buyer requirements, SOPs, market data, policies, and previous transaction context

15.2 Vision Model Requirements

The vision model should identify:

- Crop category.
- Approximate maturity/ripeness.
- Surface defects.
- Color distribution.
- Bruising.
- Mold or decay indicators.
- Size consistency.
- Packaging quality.
- Crate versus sack versus loose produce.
- Image quality and whether another image is required.

The model must not claim laboratory-grade quality certification. It should communicate:

"This is a visual estimate from the uploaded images. Final buyer acceptance may vary after p

15.3 LLM Agent Requirements

The agent must:

- Understand Urdu, Roman Urdu, and English.
- Handle short informal messages.
- Avoid requesting unnecessary information.
- Use structured tool calls instead of inventing market data.
- Distinguish between facts, predictions, and assumptions.
- Explain uncertainty.
- Ask approval before messaging or booking.
- Preserve context within an active produce lot.
- Never falsely state that a booking or sale is confirmed.
- Provide actionable next steps.

15.4 Retrieval-Augmented Generation

The agent should use RAG for:

- Crop-specific storage recommendations.
- Buyer requirement documents.
- Grade definitions.
- Packaging instructions.
- Partner policies.
- Transport and storage SOPs.
- Frequently asked user questions.
- Marketplace rules.
- Government and certification documentation guidance.

The retrieval system should never be used as the sole source of time-sensitive market prices. Prices must come from the market intelligence service with timestamp and freshness metadata.

16. AI Model Training Strategy

16.1 Initial Approach

The first version should use foundation models and rules rather than attempting to train every model from zero.

Recommended initial approach:

- Use Gemini for image understanding and multilingual conversational interaction.
- Use structured prompting and function calling for lot extraction.
- Use rule-based spoilage models initially.

- Train simple tabular machine-learning models as transaction data accumulates.
- Add human review for low-confidence visual assessments and high-value lots.

Gemini supports multimodal input, including image understanding tasks such as classification, visual question answering, and object detection. [business.whatsapp](#)

16.2 Training Data Requirements

Produce Image Dataset

For every image, collect:

Crop type
 Variety
 Location
 Harvest date
 Image timestamp
 Lighting condition
 Packaging type
 Visible defects
 Actual grade
 Buyer acceptance result
 Spoilage outcome
 Final sale price

Transaction Dataset

For every completed lot, collect:

Lot ID
 Crop
 Variety
 Quantity
 Grade
 Origin
 Destination
 Harvest date
 Packing type
 Storage duration
 Transport type
 Route duration
 Temperature exposure
 Market price at decision time
 Final selling price
 Buyer type
 Buyer acceptance

Rejected quantity
Spoilage quantity
Payment duration
Transaction completion status

Logistics Dataset

For every transport booking, collect:

Vehicle type
Refrigerated/non-refrigerated
Capacity
Pickup time
Actual pickup time
Expected delivery time
Actual delivery time
Route
Distance
Delay reasons
Temperature data if available
Driver reliability
Vehicle reliability
Delivery proof

16.3 Model Training Phases

Phase 1: Foundation and Rules

- Gemini vision for image understanding.
- LLM extraction and conversational workflows.
- Crop-specific rule tables for estimated spoilage.
- Rule-based buyer and transporter matching.
- Manual market data upload.
- Human operations review.

Phase 2: Supervised Models

Train models for:

- Crop quality grade prediction.
- Spoilage probability.
- Expected rejection probability.
- Buyer acceptance likelihood.
- Transporter reliability.
- Price movement prediction.

Phase 3: Optimization Models

Build models for:

- Net-revenue optimization.
- Dynamic market routing.
- Storage duration optimization.
- Buyer-offer ranking.
- Supply-demand forecasting.
- Route and load consolidation.

Phase 4: Continuous Learning

- Feed delivery outcome data back into models.
- Compare predicted versus actual spoilage.
- Compare recommended versus actual net revenue.
- Retrain seasonally and by region.
- Maintain separate crop and geography performance reports.

16.4 Human-in-the-Loop Process

Human review is mandatory when:

- Vision confidence is below a configurable threshold.
 - Lot value exceeds a configured threshold.
 - Buyer requirements are strict or export-related.
 - User disputes the recommendation.
 - Price data is stale or conflicting.
 - The system detects possible fraud.
 - There is a large quality mismatch between photo estimate and buyer feedback.
-

17. Spoilage Prediction Engine

17.1 Purpose

Estimate the chance and expected amount of produce spoilage under different sale, storage, and transport scenarios.

17.2 Input Variables

- Crop type.
- Variety.
- Harvest date and time.
- Ripeness.
- Visible quality.

- Packaging type.
- Current temperature.
- Forecast temperature.
- Humidity where available.
- Storage availability.
- Storage temperature.
- Transport type.
- Transport duration.
- Route delays.
- Handling quality.
- Distance.
- Buyer delivery time.
- Historical spoilage data.

17.3 Output

```
{
  "spoilage_risk_score": 0.38,
  "estimated_loss_percentage": 0.07,
  "estimated_loss_kg": 56,
  "confidence": 0.72,
  "risk_drivers": [
    "Medium-high ripeness",
    "Expected daytime temperature above recommended range",
    "Non-refrigerated transport",
    "8-hour expected wait before delivery"
  ],
  "recommended_mitigation": [
    "Use ventilated crates",
    "Dispatch before 8 AM",
    "Avoid storing longer than 24 hours",
    "Keep produce shaded before pickup"
  ]
}
```

17.4 Initial Rules

The MVP can use crop-specific rules such as:

```
If tomato ripeness is high
and ambient temperature is high
and storage is unavailable
and expected delivery is delayed,
then increase spoilage risk.
```

```
If produce is in sacks rather than ventilated crates,
```

then increase bruising and heat-retention risk.

If a cold-storage facility is available
and expected market-price increase exceeds storage cost plus expected loss,
then compare storage as a viable option.

These rules should be validated with agricultural experts before deployment.

18. Revenue Comparison Engine

18.1 Required Calculations

For each option, calculate:

- Gross sale value.
- Expected accepted quantity.
- Expected rejected quantity.
- Estimated spoilage quantity.
- Transport cost.
- Storage cost.
- Packaging cost.
- Labor/loading cost.
- Market commission.
- Platform fee.
- Payment delay risk.
- Expected net revenue.
- Confidence score.

18.2 Example Formula

[Expected Net Revenue = (Buyer Price × Expected Accepted Quantity) -
Transport Cost - Storage Cost - Commission - Packaging Cost - Handling Cost
- Platform Fee]

Where:

[Expected Accepted Quantity = Original Quantity × (1 - Estimated Spoilage Rate
- Estimated Rejection Rate)]

18.3 Recommendation Display

The user should see simple language:

Option A: Sell locally today
Expected net earnings: PKR 78,000
Risk: Low
Payment: Same day

Option B: Send to Lahore tomorrow
Expected net earnings: PKR 88,500
Risk: Medium
Payment: 2-3 days
Why: Higher price, but additional transport and spoilage risk

Recommended: Option B
Estimated benefit over local sale: PKR 10,500

19. Marketplace Design

19.1 Buyer Marketplace

Buyer profiles must include:

- Buyer name.
- Buyer category.
- Location.
- Crops purchased.
- Grade preferences.
- Minimum and maximum quantity.
- Delivery windows.
- Payment terms.
- Historical acceptance rate.
- Response rate.
- Cancellation rate.
- Ratings.
- Verification status.
- Documents required.

19.2 Transport Marketplace

Transporter profiles must include:

- Vehicle type.
- Capacity.
- Refrigerated/non-refrigerated.
- Covered/open vehicle.
- Service regions.
- Route preferences.
- Current location.
- Availability window.
- Quote method.
- Rating.

- On-time percentage.
- Cancellation rate.
- GPS capability.
- Insurance and verification documents.

19.3 Storage Marketplace

Storage profiles must include:

- Facility location.
- Total capacity.
- Available capacity.
- Temperature ranges.
- Supported crops.
- Daily cost.
- Handling fees.
- Minimum booking duration.
- Operating hours.
- Loading/unloading support.
- Verification status.
- Ratings.
- Crop-specific restrictions.

20. Functional Requirements

20.1 Authentication and Onboarding

Requirement ID	Requirement
FR-001	Users must be able to register using a Pakistani mobile number
FR-002	Users must be able to authenticate through OTP
FR-003	Users must select a role during onboarding
FR-004	Users must be able to switch or add business roles where authorized
FR-005	The system must collect consent for data processing and messaging
FR-006	Partners must complete verification before receiving marketplace priority

20.2 Produce Lot Creation

Requirement ID	Requirement
FR-010	Users must create a produce lot through photo, text, voice, or structured form input
FR-011	The system must support quantity in kg, maund, crate, bag, and ton
FR-012	The system must normalize quantities into kilograms internally
FR-013	Users must specify or confirm origin location
FR-014	Users must specify harvest readiness time
FR-015	The system must allow multiple photos per lot
FR-016	Users must be able to edit lot details before sending buyer inquiries
FR-017	The system must maintain a complete audit trail of changes

20.3 AI Analysis

Requirement ID	Requirement
FR-020	The system must identify likely crop type from photos
FR-021	The system must estimate visible quality indicators
FR-022	The system must display confidence levels for AI output
FR-023	The system must allow users to override AI-estimated crop or grade
FR-024	The agent must ask for clarification when critical information is missing
FR-025	The system must flag low-confidence assessments for review

20.4 Market Intelligence

Requirement ID	Requirement
FR-030	The system must display prices by market, crop, grade, and timestamp

Requirement ID	Requirement
FR-031	The system must distinguish live, delayed, estimated, and user-submitted prices
FR-032	The system must show price trends where historical data exists
FR-033	The system must calculate net revenue rather than only headline price
FR-034	The system must support uploaded/simulated market datasets during MVP

20.5 Buyer Matching

Requirement ID	Requirement
FR-040	The system must match lots to eligible buyers
FR-041	Matching must account for crop, grade, quantity, location, timing, and buyer requirements
FR-042	The system must display buyer reliability metrics
FR-043	The system must allow users to select one or multiple buyers for outreach
FR-044	The system must prevent buyer identity disclosure before required verification where marketplace policy requires it

20.6 Transport and Storage

Requirement ID	Requirement
FR-050	The system must find transporters matching load capacity and route
FR-051	The system must distinguish refrigerated and non-refrigerated options
FR-052	The system must calculate transport cost in the net-revenue scenario
FR-053	The system must search available storage capacity

Requirement ID	Requirement
FR-054	The system must display storage price, duration, temperature range, and location
FR-055	The system must support booking requests only after approval

20.7 Messaging and Approval

Requirement ID	Requirement
FR-060	The system must draft buyer and provider messages
FR-061	The user must explicitly approve outbound commercial messages
FR-062	The system must preserve the exact approved message content
FR-063	The system must track message delivery and response status
FR-064	The system must support WhatsApp, SMS, push, and email channels
FR-065	The system must use approved messaging templates where required

20.8 Order Management

Requirement ID	Requirement
FR-070	The system must create an order after buyer and seller confirmation
FR-071	The system must support partial quantity acceptance
FR-072	The system must support cancellation and reason capture
FR-073	The system must track pickup, transit, delivery, and payment states
FR-074	The system must store proof-of-delivery images and documents
FR-075	The system must allow disputes to be opened and reviewed

20.9 Notifications

Requirement ID	Requirement
FR-080	The system must notify users of buyer responses
FR-081	The system must notify users of transporter acceptance or decline
FR-082	The system must alert users about delivery delays
FR-083	The system must alert users when price movement changes recommendation materially
FR-084	The system must support reminder escalation for unresponsive counterparties
FR-085	Users must be able to configure notification preferences

21. Non-Functional Requirements

21.1 Performance

Requirement	Target
Basic lot intake response	Under 3 seconds excluding media upload
Image analysis initial response	Under 30 seconds under normal load
Recommendation generation	Under 90 seconds for standard scenarios
Buyer/transport matching	Under 15 seconds
Notification dispatch	Under 10 seconds after event creation
App first meaningful load	Under 4 seconds on moderate mobile network
API p95 latency	Under 500 ms for standard read endpoints

21.2 Availability

System	Target Availability
Authentication	99.9%
Lot intake	99.5%
Messaging queue	99.5%

System	Target Availability
Recommendation service	99.0%
Admin portal	99.0%
Analytics dashboards	98.5%

21.3 Scalability

The system must support:

- At least 10,000 monthly active users in pilot-ready architecture.
- At least 100,000 produce lots per month in scale-ready architecture.
- Seasonal traffic surges.
- Asynchronous processing for image analysis and bookings.
- Horizontal scaling of stateless API services.
- Queue-based handling of long-running tasks.

21.4 Security

- Encrypt data in transit using TLS.
- Encrypt sensitive data at rest.
- Use secure OTP authentication.
- Store secrets in a managed secret vault.
- Apply least-privilege access control.
- Log administrative actions.
- Sign and validate webhooks.
- Rate-limit public APIs.
- Protect against injection, SSRF, IDOR, and common OWASP risks.
- Conduct periodic security reviews and penetration testing.

21.5 Privacy

- Collect only necessary user and transaction data.
- Obtain consent for communication and location data.
- Allow users to request deletion where legally and operationally appropriate.
- Limit exact farm-location visibility according to marketplace policy.
- Redact personal information from model-training datasets where feasible.
- Keep separate permissions for user-facing and internal operational data.

21.6 Accessibility and Usability

- Support Urdu and Roman Urdu.
- Provide voice-first intake.
- Use large buttons and simple language.
- Support low-bandwidth image compression.
- Provide offline draft creation.

- Minimize typing requirements.
 - Use clear explanations instead of technical AI terminology.
-

22. Database Design

22.1 Core Entities

User
Organization
Farm
ProduceLot
LotMedia
QualityAssessment
MarketPrice
BuyerRequirement
BuyerOffer
Transporter
Vehicle
TransportQuote
StorageFacility
StorageQuote
Booking
Order
Shipment
DeliveryProof
PaymentRecord
Dispute
Notification
AgentConversation
AgentAction
ApprovalRecord
AuditLog

22.2 Example Produce Lot Schema

```
CREATE TABLE produce_lots (  
  id UUID PRIMARY KEY,  
  seller_id UUID NOT NULL,  
  organization_id UUID,  
  crop_type VARCHAR(100) NOT NULL,  
  crop_variety VARCHAR(100),  
  quantity_kg DECIMAL(12,2) NOT NULL,  
  estimated_grade VARCHAR(50),  
  seller_confirmed_grade VARCHAR(50),
```

```

origin_latitude DECIMAL(10,7),
origin_longitude DECIMAL(10,7),
origin_city VARCHAR(100),
harvest_ready_at TIMESTAMP,
storage_available BOOLEAN DEFAULT FALSE,
packaging_type VARCHAR(100),
lot_status VARCHAR(50) NOT NULL,
ai_confidence_score DECIMAL(5,4),
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL
);

```

22.3 Example Order Schema

```

CREATE TABLE orders (
    id UUID PRIMARY KEY,
    produce_lot_id UUID NOT NULL,
    seller_id UUID NOT NULL,
    buyer_id UUID NOT NULL,
    agreed_quantity_kg DECIMAL(12,2) NOT NULL,
    agreed_price_per_kg DECIMAL(12,2) NOT NULL,
    estimated_total_value DECIMAL(14,2),
    final_total_value DECIMAL(14,2),
    currency VARCHAR(10) DEFAULT 'PKR',
    order_status VARCHAR(50) NOT NULL,
    payment_status VARCHAR(50) NOT NULL,
    pickup_at TIMESTAMP,
    delivered_at TIMESTAMP,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

```

22.4 Audit Log Requirements

Every significant event must be logged:

- Who performed the action
- What action was performed
- When it happened
- What object was affected
- Previous value
- New value
- Whether the action was AI-generated
- Whether user approval was obtained
- Associated transaction ID

23. API Design

23.1 Core API Endpoints

Lot Management

```
POST /v1/lots
GET /v1/lots/{lot_id}
PATCH /v1/lots/{lot_id}
POST /v1/lots/{lot_id}/media
POST /v1/lots/{lot_id}/analyze
GET /v1/lots/{lot_id}/recommendations
```

Buyer Matching

```
GET /v1/lots/{lot_id}/buyer-matches
POST /v1/lots/{lot_id}/buyer-outreach/draft
POST /v1/lots/{lot_id}/buyer-outreach/approve
GET /v1/lots/{lot_id}/offers
```

Logistics

```
GET /v1/lots/{lot_id}/transport-matches
GET /v1/lots/{lot_id}/storage-matches
POST /v1/transport-bookings/draft
POST /v1/transport-bookings/approve
POST /v1/storage-bookings/draft
POST /v1/storage-bookings/approve
```

Orders

```
POST /v1/orders
GET /v1/orders/{order_id}
PATCH /v1/orders/{order_id}/status
POST /v1/orders/{order_id}/delivery-proof
POST /v1/orders/{order_id}/disputes
```

Agent

```
POST /v1/agent/messages
GET /v1/agent/conversations/{conversation_id}
POST /v1/agent/actions/{action_id}/approve
POST /v1/agent/actions/{action_id}/reject
```

24. WhatsApp Integration

24.1 WhatsApp Use Cases

- Receive farmer text, image, video, and voice-note inputs.
- Ask clarification questions.
- Send recommendations.
- Send approval prompts.
- Send buyer inquiries.
- Send transporter and storage booking requests.
- Send pickup and delivery notifications.
- Handle buyer confirmations and counteroffers.
- Escalate exceptions.

WhatsApp Business Platform is designed for programmatic business messaging and can support the communication layer required for this workflow. breccorder

24.2 Example WhatsApp Conversation

Farmer:

I have 800 kg tomatoes in Multan. Ready tomorrow.

FreshRoute:

Please send 2-3 photos of the tomatoes. Also tell me:

1. Are they in crates or sacks?
2. Can they leave before 9 AM tomorrow?
3. Do you have storage available tonight?

Farmer:

[Uploads photos]

They are in crates. No storage. Can leave 7 AM.

FreshRoute:

I found 3 selling options. The best expected return is a buyer in Lahore.

Expected net earnings: PKR 88,500

Estimated spoilage risk: 7%

Transport cost: PKR 9,000

Buyer reliability: High

Would you like me to contact this buyer and request transport quotes?

[Yes, contact] [Show other options]

24.3 Approval Requirement

Outbound actions with commercial consequences must require explicit approval:

- Buyer offer.
 - Counteroffer.
 - Transport booking.
 - Storage booking.
 - Cancellation.
 - Price acceptance.
 - Contract/document submission.
 - Payment request.
-

25. Admin Portal Requirements

25.1 Admin Dashboard

The admin dashboard must show:

- Active produce lots.
- Lots requiring human review.
- Pending buyer offers.
- Pending transporter requests.
- Delayed shipments.
- Buyer and transporter cancellation rates.
- Market-data freshness.
- System health.
- Revenue metrics.
- Top crops and regions.
- Disputes and escalations.

25.2 Operations Workflow

```

New low-confidence lot
    ↓
Operations queue
    ↓
Human reviewer verifies crop / quality / details
    ↓
Reviewer corrects data or contacts user
    ↓
Lot returns to recommendation flow

```

25.3 Partner Verification

Admin users must be able to verify:

- Buyer business identity.
- Transporter identity and vehicle documents.

- Cold-storage facility documents.
 - Bank/payment details where applicable.
 - Trade history.
 - Operational location.
-

26. Trust, Safety, and Fraud Prevention

26.1 Fraud Risks

- Fake produce listings.
- Fake buyer offers.
- Fake transporter availability.
- Price manipulation.
- Quantity fraud.
- Payment fraud.
- Delivery proof fraud.
- Fake storage capacity.
- Misrepresentation of crop quality.

26.2 Mitigation Measures

- Phone and identity verification.
 - Partner verification badges.
 - Document verification.
 - Transaction history.
 - Ratings and reliability scores.
 - GPS and timestamp validation where possible.
 - Image metadata analysis where legally appropriate.
 - Proof-of-delivery workflows.
 - Escrow or payment-partner integration in future versions.
 - Risk scoring for abnormal behavior.
 - Human review for high-value transactions.
-

27. Metrics and KPIs

27.1 North Star Metric

Incremental net value realized per completed produce lot through FreshRoute.

27.2 User Metrics

- Monthly active farmers.
- Monthly active traders.
- Produce lots created.
- Repeat lot creation rate.
- Recommendation acceptance rate.
- Buyer outreach approval rate.
- Booking completion rate.
- User retention by crop season.
- WhatsApp-to-app conversion rate.

27.3 Economic Metrics

- Gross merchandise value.
- Platform revenue.
- Average transaction value.
- Average commission per transaction.
- Net revenue uplift compared with local-market baseline.
- Estimated spoilage avoided.
- Cold-storage utilization.
- Transport fill rate.
- Buyer fill rate.

27.4 Operational Metrics

- Time from lot creation to recommendation.
- Time from recommendation to buyer response.
- Time from buyer confirmation to pickup.
- Delivery success rate.
- On-time delivery rate.
- Buyer cancellation rate.
- Transporter cancellation rate.
- Dispute rate.
- Human-review rate.

27.5 AI Metrics

- Crop-recognition accuracy.
- Quality-estimation agreement with human graders.
- Spoilage-prediction calibration.
- Price-forecast error.
- Recommendation acceptance rate.
- Recommendation-to-success conversion rate.
- Hallucination rate.
- Tool-call success rate.

- Low-confidence escalation rate.

28. Business Model

28.1 Revenue Streams

Revenue Stream	Description
Transaction commission	Percentage fee on completed produce transactions
Transport booking fee	Commission on successful transporter bookings
Storage booking fee	Commission on successful cold-storage bookings
SaaS subscriptions	Monthly plans for traders, buyers, collection centers, and enterprises
Premium analytics	Market intelligence, price forecasting, supply visibility, and operational dashboards
Enterprise API	APIs for banks, insurers, agri-input firms, cooperatives, and procurement companies
Verification services	Paid verification or premium profile for buyers, transporters, and storage providers
Finance referral	Referral fee for working-capital, invoice-finance, or insurance partners
Featured buyer demand	Paid promotion for verified buyer demand requests

28.2 Suggested Pricing Model

Farmers

- Free basic market and advisory access.
- Low commission only on successfully completed agent-assisted transactions.
- Optional premium services for priority matching or on-call operations support.

Traders and Collection Centers

- Monthly subscription.
- Per-transaction commission.
- Team dashboard and multi-user access.

- Premium analytics package.

Buyers

- Free initial onboarding.
- Subscription for procurement dashboard.
- Buyer posting fee or successful sourcing fee.
- Premium supplier intelligence.

Transporters and Storage Providers

- Free basic listing.
- Commission per completed booking.
- Premium ranking/visibility package.
- Fleet and capacity management subscription.

28.3 Illustrative Unit Economics

Average produce transaction value: PKR 150,000

Platform transaction commission: 1.5%

Revenue per transaction: PKR 2,250

Transport booking value: PKR 15,000

Transport commission: 5%

Revenue per transport booking: PKR 750

Total potential revenue for one completed transaction:

PKR 3,000+ before subscription and analytics revenue

The commission should be justified by measurable user value, such as higher realized price, lower spoilage, lower coordination time, or more reliable buyer access.

29. Go-To-Market Strategy

29.1 Launch Strategy

Start with a narrow, controlled marketplace:

- One crop cluster.
- One or two farm-to-market corridors.
- Verified buyers.
- Verified transporters.
- Selected cold-storage providers.
- Dedicated human operations team.
- Manual market-data validation.

29.2 Pilot Proposal

Pilot Crop

Tomatoes.

Pilot Location

Multan to Lahore and Multan to Islamabad/Rawalpindi.

Pilot Users

- 50 farmers.
- 10 traders/collection centers.
- 20 buyers.
- 20 transporters.
- 5 storage facilities.

Pilot Duration

12 weeks.

Pilot Success Criteria

- 500+ produce lots created.
- 100+ completed transactions.
- 20%+ recommendation-to-transaction conversion.
- 10%+ average net-revenue improvement versus the user's baseline.
- 15%+ reduction in estimated spoilage for participating lots.
- 70%+ user satisfaction among active users.

29.3 Acquisition Channels

- Mandi commission agents.
 - Agricultural input dealers.
 - Farmer organizations and cooperatives.
 - Collection centers.
 - WhatsApp referral loops.
 - Agricultural extension programs.
 - NGOs and development programs.
 - Buyer procurement partnerships.
 - Cold-chain and logistics partners.
-

30. Product Roadmap

Phase 0: Discovery and Validation

Duration: 4–6 weeks.

- Interview farmers, traders, buyers, transporters, and storage providers.
- Map actual selling workflows.
- Identify top crops and corridors.
- Validate willingness to pay.
- Gather initial market-price data.
- Build partner pipeline.
- Define quality taxonomy.

Phase 1: MVP

Duration: 8–12 weeks.

- User onboarding.
- Produce-lot creation.
- Photo upload.
- Basic Gemini-based crop and visible-quality analysis.
- Text/voice intake.
- Manual/uploaded market-price data.
- Sell-now versus store comparison.
- Buyer matching.
- Transporter matching.
- WhatsApp communication.
- User approval workflow.
- Basic order tracking.
- Admin portal.

Phase 2: Pilot Marketplace

Duration: 3–6 months.

- Verified buyer network.
- Transport booking.
- Cold-storage booking.
- Live market feeds.
- Price trends.
- Reliability scores.
- Driver tracking.
- Buyer counteroffers.
- Dispute handling.
- Operational analytics.

Phase 3: Intelligence and Automation

Duration: 6–12 months.

- Crop-specific spoilage models.
- Price forecasting.
- Dynamic routing.
- Load consolidation.
- Automated re-planning.
- Payment integrations.
- Digital quality certificates.
- Finance and insurance integrations.
- Enterprise APIs.

Phase 4: Platform Expansion

Duration: 12+ months.

- More crops.
 - More regions.
 - Export workflows.
 - Traceability.
 - Farm-level supply forecasting.
 - Carbon and waste analytics.
 - Regional cross-border trade tools.
-

31. MVP Acceptance Criteria

The MVP is ready for pilot when:

- A user can create a produce lot from text and at least one photo.
 - The system can identify a supported crop with confidence scoring.
 - The system can ask follow-up questions.
 - The system can create a sell-now versus store comparison.
 - The system can show market price data with timestamps.
 - The system can recommend at least one buyer and one transport option.
 - The system can draft WhatsApp outreach messages.
 - No buyer message or booking can occur without user approval.
 - The system can track a transaction from inquiry through delivery.
 - Admin users can review low-confidence cases.
 - Every agent action and approval is auditable.
 - Basic analytics report lots, recommendations, outreach, bookings, and completed transactions.
-

32. Risks and Mitigations

Risk	Impact	Mitigation
Inaccurate market prices	Poor recommendations and loss of trust	Display source, timestamp, confidence, and freshness; use multiple sources; allow manual validation
AI quality misclassification	Buyer rejection and user dissatisfaction	Confidence thresholds, human review, user override, clear visual-estimate disclaimer
Low user trust	Weak adoption	Explain recommendations, show net-revenue breakdown, use local partners, enable human support
Transport unreliability	Delivery failure	Ratings, verification, backup transporter matching, penalties and cancellation tracking
Buyer cancellation	Revenue loss and spoilage	Buyer reliability scores, deposits in later phases, rapid rematching workflow
Low smartphone literacy	Poor product adoption	WhatsApp-first workflow, voice input, simple UI, field-agent support
Incomplete data	Weak predictions	Start with rules and operational data collection; improve models over time
Seasonal market volatility	Recommendation errors	Real-time data refresh, uncertainty display, alert-based re-planning
Fraud	Financial and reputational risk	Verification, audit logs, reliability scoring, manual review, proof of delivery

Regulatory or policy limitations	Messaging disruption	Follow messaging platform policy, use approved templates, maintain SMS fallback
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33. Why Judges Will Remember FreshRoute Agent

FreshRoute Agent stands out because it is not another agricultural information chatbot.

It is an **execution agent** that transforms a simple farmer message into a complete commercial workflow:

```

Photo / voice note / text
↓
AI quality analysis
↓
Market and revenue comparison
↓
Spoilage-risk estimate
↓
Buyer, transport, and storage matching
↓
User-approved outreach
↓
Booking and coordination
↓
Delivery tracking
↓
Real-time alerts and re-planning

```

The product is memorable because it connects multiple disconnected problems—quality assessment, market intelligence, logistics, storage, buyer coordination, and transaction monitoring—into one practical workflow.

It has a clear social and commercial outcome:

Better farmer income, lower produce waste, faster market access, and more efficient agricultural logistics.

The use of multimodal AI is practical rather than decorative: Gemini can interpret produce photos and support conversational intake, while the rest of the system converts that understanding into real business decisions and controlled actions. `business.whatsapp`

34. Final Product Positioning

FreshRoute Agent is an AI-powered selling and logistics copilot for fresh produce. It helps farmers and traders decide where, when, and how to sell—and then executes the buyer, transport, storage, tracking, and alert workflow with user approval.

It is the intelligence and execution layer between the farm, mandi, buyer, transporter, and cold-storage ecosystem.